Introduction to SQL & Relational Databases

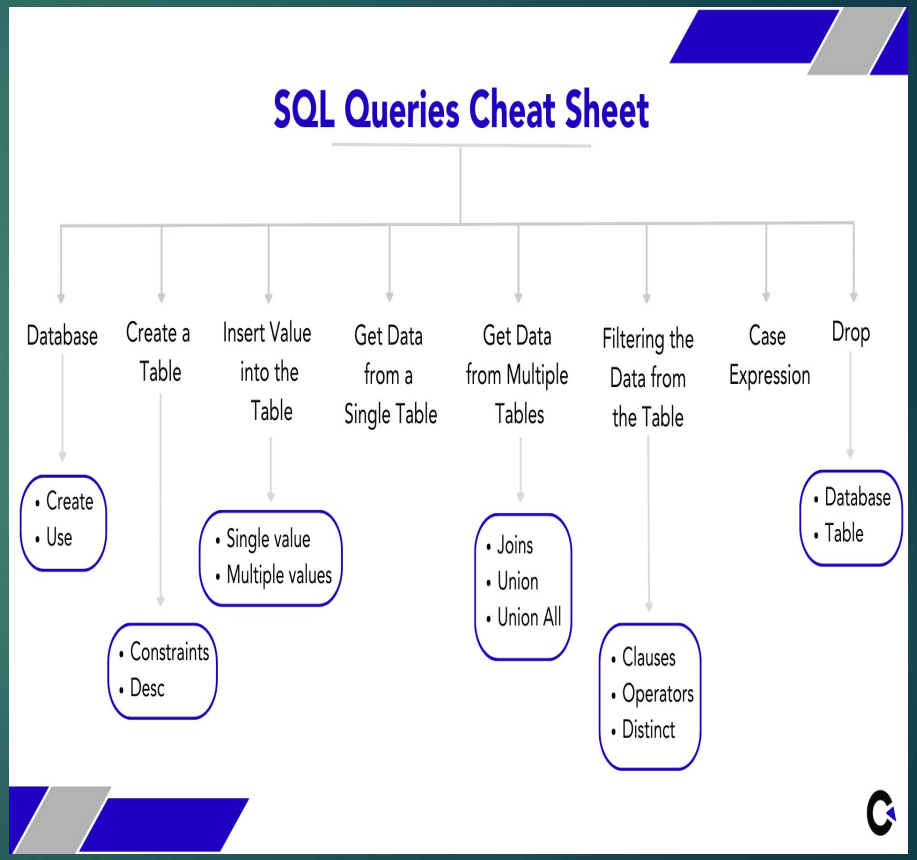
STRUCTURED QUERY LANGUAGE (SQL) IS THE STANDARD LANGUAGE FOR MANAGING AND MANIPULATING RELATIONAL DATABASES.

RDBMS vs. SQL

Feature	RDBMS	SQL
Full Form	Relational Database Management System	Structured Query Language
Definition	The software/system that manages and stores data (e.g., MySQL, Oracle, SQL Server).	The standard programming language used to interact with the database.
Role	The **Engine**: It handles storage, security, and data integrity.	The **Interface**: It provides the syntax to query, insert, or delete data.
Analogy	The **Warehouse**: The physical space where goods are kept and organized.	The **Instructions**: The work order used to retrieve or move specific goods.
Examples	PostgreSQL, MySQL, Microsoft SQL Server, SQLite.	SELECT, INSERT, UPDATE, DELETE, JOIN.

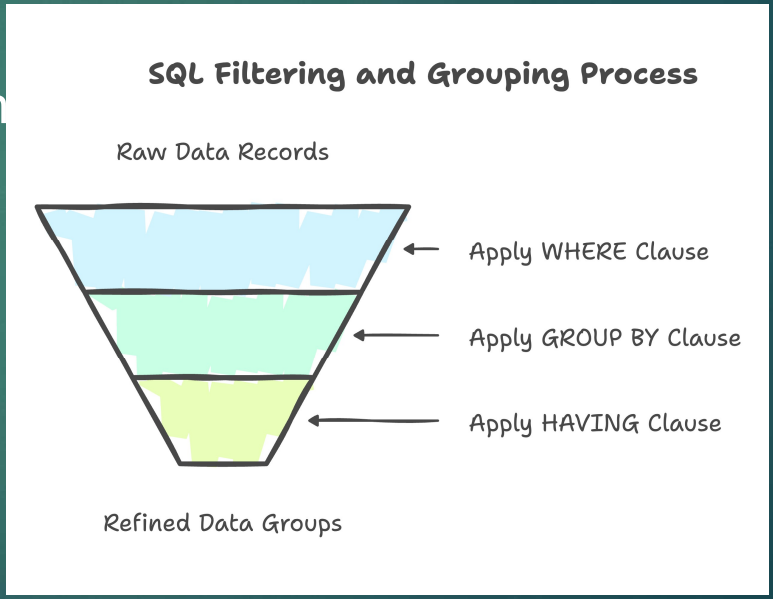
Basic Queries

- **SELECT:** Identifies which columns to retrieve.
- **FROM:** Specifies the table where the data resides.
- **DISTINCT:** Filters out duplicate values to return only unique entries.
- **AS (Aliasing):** Temporarily renames a column or table for better readability in the result set.
- **Example:** **SELECT DISTINCT City AS Location FROM Customers;**



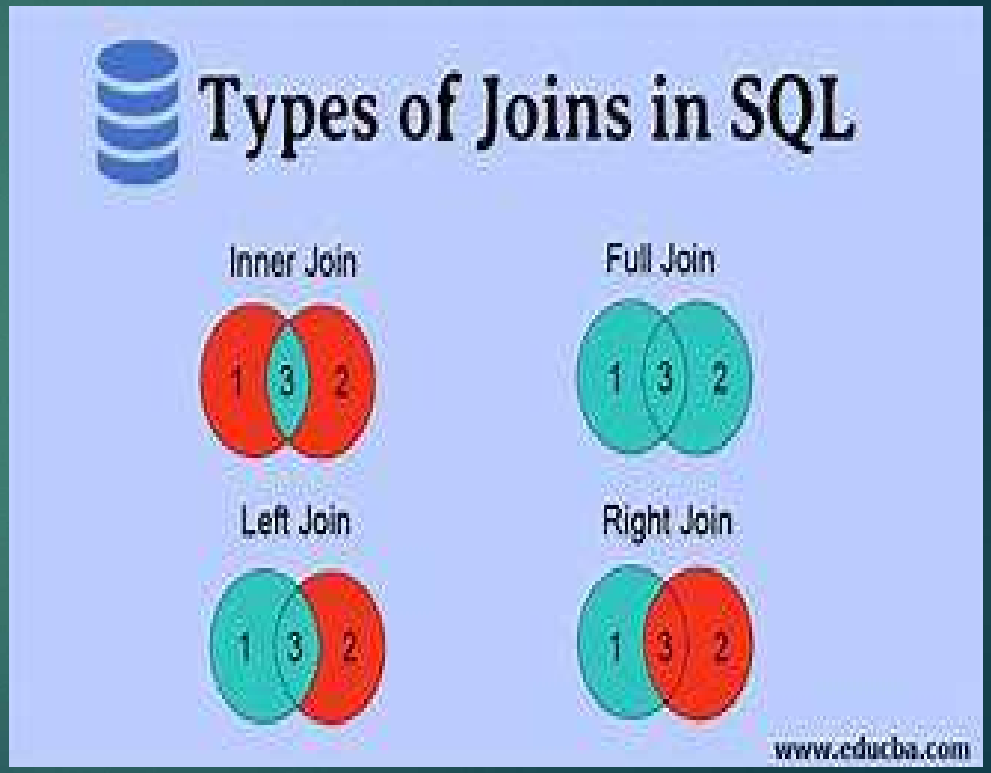
Filtering & Sorting

- **WHERE:** Filters records based on specific conditions
- **ORDER BY:** Sorts the result set in ascending
- **LIMIT:** Restricts the number of rows returned
  (useful for pagination or top-N analysis).
- **Example:** **SELECT * FROM Orders WHERE Status = 'Shipped' ORDER BY Date DESC LIMIT 5;**



Types of Joins

•**INNER JOIN:** Returns records that have matching
values in both tables.
•**LEFT (OUTER) JOIN:** Returns all records from
the left table and the matched records from the
right table.
•**RIGHT (OUTER) JOIN:** Returns all records from
 the right table and the matched records from the
 left table.
•**FULL (OUTER) JOIN:** Returns all records when
 there is a match in either the left or right table.



SQL Constraints & Data Integrity

- **NOT NULL:** Ensures a column cannot have a NULL value.
- **UNIQUE:** Ensures all values in a column are different.
- **PRIMARY KEY:** A field that uniquely identifies each record in a table..
- **FOREIGN KEY:** Prevents actions that would destroy links between tables (referential integrity
- **CHECK:** Ensures the values in a column satisfy a specific condition.
- **DEFAULT:** Sets a default value for a column if no value is specified.